

Lecture: Convergence of Gradient Descent

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In the last lecture, we established that for convex functions, any critical point where $\nabla f(x) = 0$ is a global minimum. Today, we will show that gradient descent converges to the global minima for smooth convex functions.

1 Smoothness (Lipschitz Gradient)

While convexity prevents a function from having multiple disconnected minima, it doesn't prevent the function's curvature from being arbitrarily large. An algorithm can easily overshoot and diverge if the gradient changes too rapidly. *Smoothness* is a condition that limits how fast the gradient can change.

Definition 1 (Smoothness). *A function f is L -smooth for some $L > 0$ if its gradient is Lipschitz continuous with constant L . Formally, for all x, y :*

$$\|\nabla f(x) - \nabla f(y)\|_2 \leq L\|x - y\|_2$$

This definition means the gradient cannot change arbitrarily fast. For twice-differentiable functions, smoothness is equivalent to having an upper bound on the Hessian, which means the function's curvature cannot be infinite.

Property 2 (Second-Order Condition for Smoothness). *A twice-differentiable function f is L -smooth if and only if its Hessian is bounded above:*

$$\nabla^2 f(x) \preceq LI \quad \text{for all } x$$

This means the largest eigenvalue of the Hessian is at most L .

An L -smooth function is quadratically upper-bounded. This is a crucial consequence that we will use in our proof.

Lemma 3 (Quadratic Upper Bound). *If f is L -smooth, then for all x, y :*

$$f(y) \leq f(x) + \nabla f(x)^\top (y - x) + \frac{L}{2} \|y - x\|_2^2$$

This inequality says that the function doesn't curve *upwards* too much faster than a quadratic with curvature L . It provides a global upper bound on the function's value.

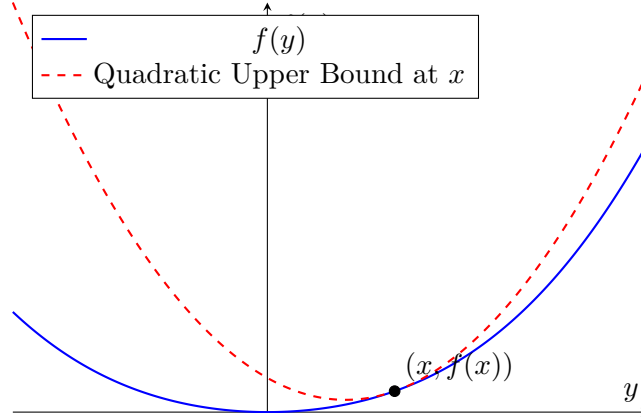


Figure 1: An L -smooth function is upper-bounded by a quadratic function at every point.

2 Convergence of Gradient Descent for Convex Functions

Recall the gradient descent algorithm:

Algorithm 1: Gradient Descent

Input: Starting point x_0 , step size η , number of iterations T

Output: Final parameters x_T

Initialize x_0 ;

for $t = 0, \dots, T - 1$ **do**

$x_{t+1} = x_t - \eta \nabla f(x_t)$;

end

return x_T ;

We have all the pieces to prove the convergence of gradient descent for functions that are both convex and smooth. We will prove the following theorem:

Theorem 4. *Let f be a convex and L -smooth function. If we run gradient descent with a fixed step size $\eta \leq 1/L$, then for any optimal point x^* :*

$$f(x_T) - f(x^*) \leq \frac{\|x_0 - x^*\|_2^2}{2\eta T}$$

This result shows that the “optimality gap” $f(x_T) - f(x^*)$ shrinks at a rate of $O(1/T)$. This means that to guarantee the error is less than some ϵ , we need to run for $T \geq \frac{\|x_0 - x^*\|_2^2}{2\eta\epsilon}$ iterations, so the number of iterations is $O(1/\epsilon)$.

2.1 Proof of Theorem 4

The proof can seem technical, but it’s built on three simple ideas that we will assemble.

1. **The Descent Lemma:** First, we’ll derive a key result known as the Descent Lemma, which shows that a small enough step ($\eta \leq 1/L$) guarantees the function value decreases.

2. **Relating Iterates to the Optimum:** Next, we'll use the convexity property to relate the optimality gap at step t to the distance from the optimum, $\|x_t - x^*\|^2$.
3. **Summing Up Progress Over Time:** Finally, we will combine the results from the first two steps and sum over all T iterations. A clever trick called a telescoping sum will make most of the terms cancel out, leaving the final $O(1/T)$ rate.

Now, let's execute this strategy.

Step 1: The Descent Lemma. First, we establish the crucial *Descent Lemma*. This lemma states that a small enough step size guarantees the function value decreases as long as gradient is non-zero.

Lemma 5 (Descent Lemma). *For an L -smooth function, a gradient descent step with $\eta \leq 1/L$ guarantees that*

$$f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2.$$

Proof. We start with the quadratic upper bound from the smoothness property, letting $x = x_t$ and $y = x_{t+1}$:

$$\begin{aligned} f(x_{t+1}) &\leq f(x_t) + \nabla f(x_t)^\top (x_{t+1} - x_t) + \frac{L}{2} \|x_{t+1} - x_t\|_2^2 && \text{(smoothness upper bound)} \\ &= f(x_t) + \nabla f(x_t)^\top (-\eta \nabla f(x_t)) + \frac{L}{2} \|\eta \nabla f(x_t)\|_2^2 && \text{(substitute } x_{t+1} = x_t - \eta \nabla f(x_t)) \\ &= f(x_t) - \eta \|\nabla f(x_t)\|_2^2 + \frac{L\eta^2}{2} \|\nabla f(x_t)\|_2^2 && \text{(simplify inner product and norm)} \\ &= f(x_t) - \eta \left(1 - \frac{L\eta}{2}\right) \|\nabla f(x_t)\|_2^2 && \text{(factor out } \eta \|\nabla f(x_t)\|_2^2) \end{aligned}$$

If we choose the step size $\eta \leq 1/L$, then $(1 - L\eta/2) \geq 1/2$, so $f(x_{t+1}) \leq f(x_t) - \frac{\eta}{2} \|\nabla f(x_t)\|_2^2$. $\square$

Step 2: Relating Iterates to the Optimum. Next, we analyze how the distance to the optimum x^* changes in one step.

$$\begin{aligned} \|x_{t+1} - x^*\|_2^2 &= \|x_t - \eta \nabla f(x_t) - x^*\|_2^2 && \text{(substitute GD update)} \\ &= \|(x_t - x^*) - \eta \nabla f(x_t)\|_2^2 && \text{(regroup terms)} \\ &= \|x_t - x^*\|_2^2 - 2\eta \nabla f(x_t)^\top (x_t - x^*) + \eta^2 \|\nabla f(x_t)\|_2^2 \\ &\quad \text{(expand } \|a - b\|^2 = \|a\|^2 - 2a^\top b + \|b\|^2) \end{aligned}$$

Using the first-order condition for convexity, $\nabla f(x_t)^\top (x_t - x^*) \geq f(x_t) - f(x^*)$, we can bound the middle term:

$$\begin{aligned} \|x_{t+1} - x^*\|_2^2 &\leq \|x_t - x^*\|_2^2 - 2\eta(f(x_t) - f(x^*)) + \eta^2 \|\nabla f(x_t)\|_2^2 \\ &\quad \text{(use convexity to lower bound middle term)} \end{aligned}$$

Rearranging this to isolate the optimality gap gives our key inequality from this step:

$$2\eta(f(x_t) - f(x^*)) \leq \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 + \eta^2 \|\nabla f(x_t)\|_2^2$$

Step 3: Summing Up Progress Over Time. The inequality from Step 2 is useful, but it still has a gradient term on the right. We can eliminate it using the Descent Lemma. With $\eta = 1/L$, the lemma gives us $\|\nabla f(x_t)\|_2^2 \leq 2L(f(x_t) - f(x_{t+1}))$. Substituting this into the result from Step 2:

$$\begin{aligned} \frac{2}{L}(f(x_t) - f(x^*)) &\leq \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 + \frac{1}{L^2} \cdot 2L(f(x_t) - f(x_{t+1})) \\ &\quad \text{(set } \eta = 1/L \text{ and substitute bound on gradient)} \\ &= \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 + \frac{2}{L}(f(x_t) - f(x_{t+1})) \\ &\quad \text{(simplify } \frac{1}{L^2} \cdot 2L = \frac{2}{L} \text{)} \end{aligned}$$

The $\frac{2}{L}f(x_t)$ terms on both sides cancel, and we can rearrange to get a clean, per-step progress inequality:

$$\begin{aligned} \frac{2}{L}f(x_{t+1}) - \frac{2}{L}f(x^*) &\leq \|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2 \\ &\quad \text{(cancel } \frac{2}{L}f(x_t) \text{ from both sides)} \\ f(x_{t+1}) - f(x^*) &\leq \frac{L}{2} (\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2) \\ &\quad \text{(multiply both sides by } \frac{L}{2} \text{)} \end{aligned}$$

Now we sum this over all iterations $t = 0, \dots, T-1$:

$$\begin{aligned} \sum_{t=0}^{T-1} (f(x_{t+1}) - f(x^*)) &\leq \frac{L}{2} \sum_{t=0}^{T-1} (\|x_t - x^*\|_2^2 - \|x_{t+1} - x^*\|_2^2) \\ &\quad \text{(sum over all } T \text{ iterations)} \\ &= \frac{L}{2} (\|x_0 - x^*\|_2^2 - \|x_T - x^*\|_2^2) \\ &\quad \text{(telescoping sum: terms cancel)} \\ &\leq \frac{L}{2} \|x_0 - x^*\|_2^2 \\ &\quad \text{(drop non-negative term)} \end{aligned}$$

Since $f(x_t)$ is non-increasing (from Step 1), we have $f(x_T) \leq f(x_t)$ for all t . This means $f(x_T) \leq f(x_{t+1})$. Therefore:

$$\begin{aligned} T(f(x_T) - f(x^*)) &\leq \sum_{t=0}^{T-1} (f(x_{t+1}) - f(x^*)) \leq \frac{L}{2} \|x_0 - x^*\|_2^2 \\ &\quad \text{(bound sum by worst term)} \end{aligned}$$

Finally, dividing by T and using $\eta = 1/L$ gives the desired result:

$$\begin{aligned} f(x_T) - f(x^*) &\leq \frac{L\|x_0 - x^*\|_2^2}{2T} = \frac{\|x_0 - x^*\|_2^2}{2\eta T} \\ &\quad \text{(divide by } T \text{ and substitute } \eta = 1/L \text{)} \end{aligned}$$

3 Faster Convergence under Strong Convexity

If we additionally assume the function is μ -strongly convex, we get a much faster rate.

Theorem 6. *Let f be μ -strongly convex and L -smooth. With $\eta = 1/L$, GD converges at a linear rate:*

$$\|x_T - x^*\|_2 \leq \left(1 - \frac{\mu}{L}\right)^T \|x_0 - x^*\|_2$$

This is called a *linear* convergence rate because the error is multiplied by a constant factor < 1 at each step, analogous to how a geometric series decreases. The error gets reduced by a constant factor every iteration, so the number of iterations to get ϵ error is $O(\log(1/\epsilon))$. This is much faster than the $O(1/\epsilon)$ rate for general convex functions.

While we will not see the general proof here, in the homework, we will show the result for the least squares problem.